ADVANCED MACHINE LEARNING PROJECT

Flight Club

Flight Delay Prediction

Project workflow: data understanding to model evaluation

This project investigates whether information available before departure can help predict whether a flight will arrive more than 15 minutes after its scheduled arrival time.

Submitted by

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Repository

https://github.com/suhanigupta786/T004-Flight-Club

Report scope: This report incorporates saved outputs from the project notebooks. Model metrics are reported as recorded in those runs; the additional calibration experiment is distinguished from the main model comparison.

Abstract

Flight Club investigates a supervised machine-learning approach for predicting whether a US domestic flight will arrive more than 15 minutes after its scheduled arrival time. The report is organized in the team's requested workflow: dataset understanding and data dictionary; exploratory data analysis and target definition; leakage audit and feature engineering; chronological split and preprocessing; Logistic Regression; XGBoost; and evaluation and comparison. The documented workflow uses 2013 historical flight records and emphasizes prediction-time validity: variables produced after departure or arrival must not be used as pre-departure model inputs. Logistic Regression provides an interpretable baseline, while XGBoost is intended to capture nonlinear relationships and interactions. The repository's README marks final model-comparison metrics as pending, so this draft does not fabricate results. The final version should insert verified metrics, plots, and conclusions from the team's executed notebooks. **Keywords:** flight delay prediction, binary classification, Logistic Regression, XGBoost, feature engineering, data leakage, chronological validation.

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Chapter 1

Introduction

1.1 Background

Flight delays can disrupt passenger journeys, increase operating costs, affect connecting flights, and contribute to airport congestion. A prediction system may help identify flights at higher risk of delay, but it must use only information that would be available at the time the prediction is made.

1.2 Problem Statement

Given information available before a flight departs, predict whether its actual arrival time will be more than 15 minutes later than its scheduled arrival time. Let A_i be actual arrival time and S_i scheduled arrival time. The arrival delay is $D_i = A_i - S_i$, and the target is

$$y_i = \begin{cases} 1, & D_i > 15 \\ 0, & D_i \leq 15. \end{cases}$$

The model estimates $P(y_i = 1 \mid X_i)$ using only valid pre-departure predictors X_i.

1.3 Project Objectives

- Understand the dataset and document the variables.
- Explore the data and define the classification target.
- Prevent target leakage and engineer valid features.
- Create chronological partitions and fit preprocessing without using future data.
- Establish a Logistic Regression baseline and train an XGBoost classifier.
- Evaluate models using metrics appropriate for a potentially imbalanced classification task.

Chapter 2

01 — Dataset Understanding and Data Dictionary

2.1 Dataset Overview

The EDA notebook begins with 336,776 2013 flight records. It identifies 8,255 cancellations and 1,175 diverted/no-arrival records, leaving 327,346 completed flights with an observed arrival-delay target. The modelling notebooks use those 327,346 rows. Predictors include scheduled times, carrier and airport identifiers, route, distance, calendar encodings, and lagged historical delay summaries.

Table 2.1: Dataset characteristics documented in the repository

Characteristic	Description
Time period	January–December 2013
Geography	United States
Documented cleaned records	327,346
Prediction task	Binary arrival-delay classification
Target	`delayed`
Positive class	Arrival delay greater than 15 minutes
Candidate categorical variables	Carrier, origin airport, destination airport, route
Candidate numeric variables	Distance, temporal encodings, historical statistics
Prediction timing	Before departure

2.2 Data Dictionary

The data dictionary should describe each field's meaning, data type, role, and whether it is valid at prediction time. The following summarizes the main field families documented by the project.

Field/family	Role	Notes
`arr_delay`	Target source	Used to construct `delayed`; must not be an input feature.
`delayed`	Target	1 when arrival delay exceeds 15 minutes; otherwise 0.
Scheduled departure/arrival time	Candidate predictor	Available before departure if represented as scheduled values.
`carrier`	Categorical predictor	Airline identifier.
`origin`, `dest`	Categorical predictors	Origin and destination airports.
`distance`	Numeric predictor	Scheduled route distance.
Calendar/time fields	Engineered predictors	Month, day, weekday, hour, and cyclic encodings where used.
Historical delay rates/counts	Engineered predictors	Must be computed only from prior dates/observations.
`dep_delay`, actual departure/arrival times	Excluded from inputs	Post-event operational information.
`air_time`	Excluded from inputs	Actual flight duration is not known before departure.
Identifiers and split labels	Metadata	Do not use as predictors unless a documented justification exists.

2.3 Data Quality Checks

In the initial 336,776-row data, 9,430 records have missing arrival delay (2.80%); the notebook classifies these as cancelled or diverted/no-arrival and excludes them from the binary target. Among completed flights, the EDA notebook reports 327,346 records, with 8,255 missing actual departure times in the original table and no duplicated exported row IDs in the modelling-stage checks. The target is never imputed: records without an observed arrival delay are not labelled as on-time.

Chapter 3

02 — Exploratory Data Analysis and Target

3.1 Exploratory Data Analysis

EDA examines class balance, the long-tailed arrival-delay distribution, seasonal and weekday variation, airline/airport differences, route distance, and day-to-day temporal variation. The completed-flight target contains 77,630 delayed flights (23.71%) and 249,716 non-delayed flights (76.29%), an approximate 3.22:1 class imbalance. The positive class is defined strictly as arrival delay greater than 15 minutes. The saved notebook figures are reproduced below.

3.2 Target Definition

The target is constructed from `arr_delay`: `delayed`=1 if `arr_delay`>15, and `delayed`=0 if `arr_delay`≤15. Rows with a missing arrival delay require explicit exclusion or a documented alternative policy. The target source must be removed from the predictor matrix.

3.3 Key EDA Figures

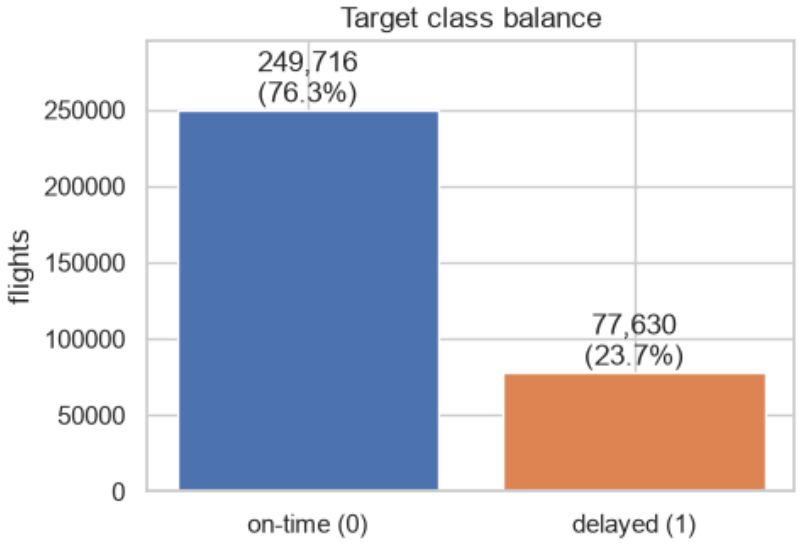


Figure 3.1: Target class balance in completed flights. The delayed class accounts for 23.71% of the modelling data.

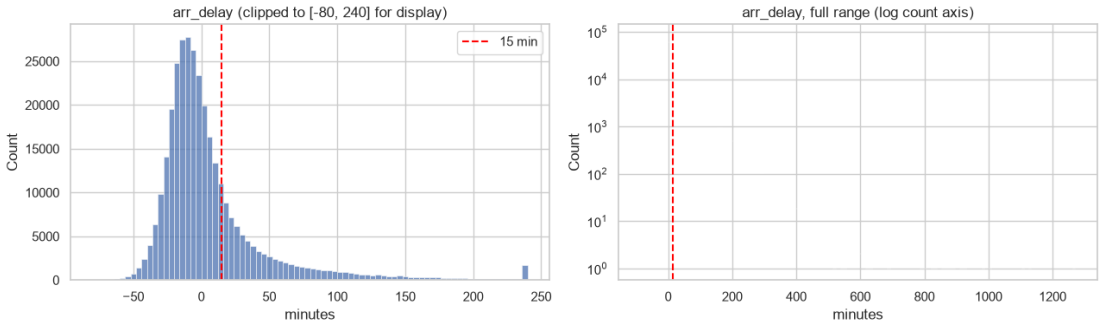


Figure 3.2: Arrival-delay distribution and delay indicator diagnostics. Arrival delays are right-skewed, with a long tail of severe delays.

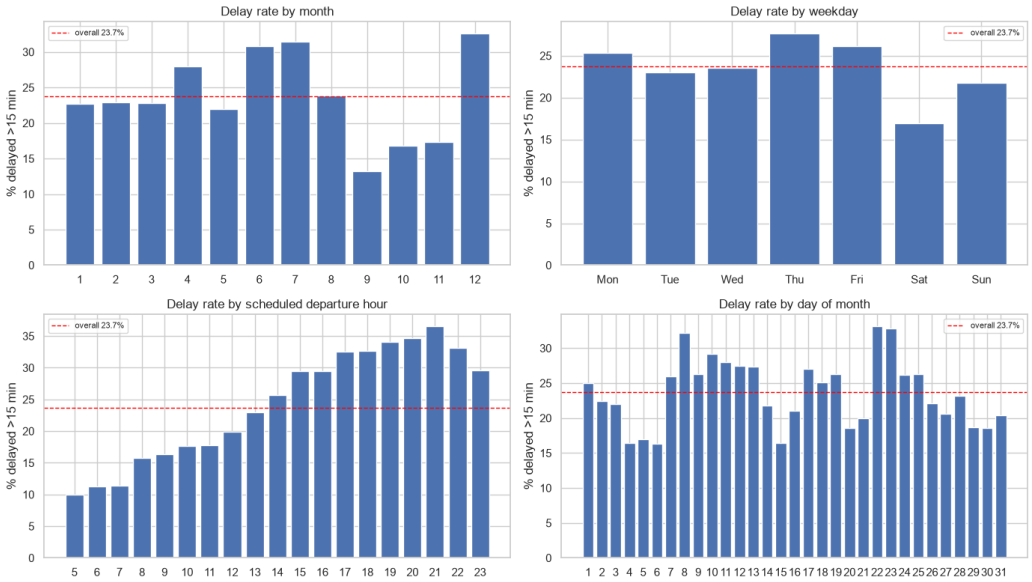


Figure 3.3: Exploratory temporal patterns by month, weekday, scheduled departure hour, and weekday/hour. These are descriptive patterns, not causal effects.

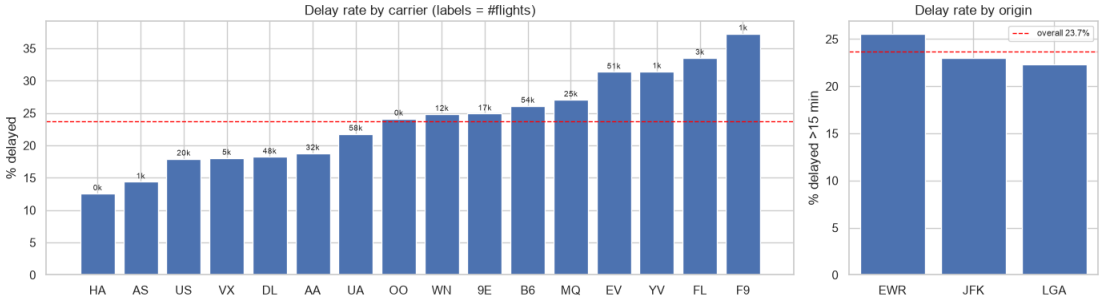


Figure 3.4: Airline-level delay-rate comparison. Rates should be interpreted alongside each carrier's sample size.

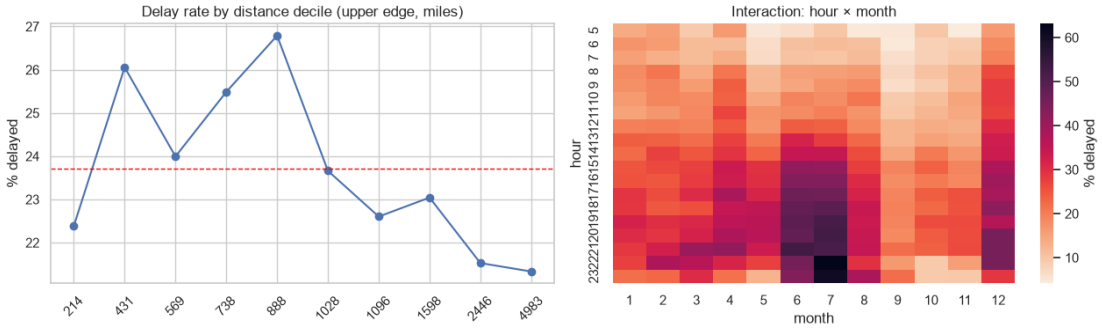


Figure 3.5: Delay-rate variation across route-distance bins and temporal groupings. Distance alone does not explain the full delay pattern.

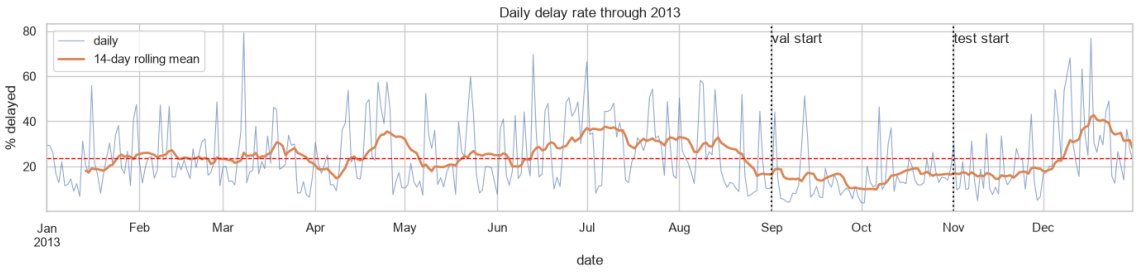


Figure 3.6: Daily delay-rate series across 2013. The visible temporal variation motivates chronological validation and historical features.

3.4 EDA Interpretation

The arrival-delay distribution is strongly right-skewed (reported skewness 3.72); its median is -5 minutes, while its 95th percentile is 91 minutes. This demonstrates why the mean alone is insufficient to summarize delay behaviour. Delay rates differ by airline, month, weekday, and scheduled departure hour, but these group-level associations do not establish causality. The notebook reports lag-1 autocorrelation of daily delay rate of 0.395 and lag-7 autocorrelation of 0.113, supporting investigation of prior-day context while also requiring careful time ordering. A separate EDA plot shows that departure delay is strongly associated with arrival delay (Pearson correlation 0.915); this is precisely why actual departure delay is excluded from a pre-departure model.

Chapter 4

03 — Leakage Audit and Feature Engineering

4.1 Leakage Audit

Data leakage occurs when a predictor contains information unavailable at the prediction time or directly reveals the target. Because the intended use is pre-departure prediction, actual operational outcomes must not be model inputs.

Field	Decision	Reason
Scheduled times	Retain/engineer	Known before departure.
Carrier, origin, destination, distance	Candidate predictors	Flight and route characteristics.
Historical delay rates/counts	Retain with safeguards	Must use only prior information.
`arr_delay`	Exclude from features	Used to construct the target.
`dep_delay`, actual departure/arrival time	Exclude	Known only after operations have progressed.
`air_time`	Exclude	Actual flight duration is not known pre-departure.
IDs and split labels	Exclude from predictors	Metadata rather than generalizable flight characteristics.

4.2 Temporal Feature Engineering

For a cyclic feature x with period T, encode it as

$$x_{\sin} = \sin\left(\frac{2\pi x}{T}\right), \qquad x_{\cos} = \cos\left(\frac{2\pi x}{T}\right).$$

This can represent cyclical patterns in scheduled time and calendar variables without treating the end and beginning of a cycle as far apart.

4.3 Route and Historical Features

A combined origin–destination identifier may represent a route as a categorical feature. Historical delay rates and flight counts may be computed for airports, carriers, and routes over a preceding seven-calendar-day window. These features must exclude the current date and any outcome not yet known at prediction time. Where no history exists, retain a count or other indicator so a zero rate is not confused with a well-supported zero-delay estimate.

4.4 Feature Audit

The final notebook should list the exact predictors used by each model and confirm that no target-derived or post-departure fields remain. Feature importance is useful for describing model behaviour but does not prove that a feature causes delays.

Chapter 5

04 — Chronological Split and Preprocessing

5.1 Chronological Data Split

The executed preprocessing and model notebooks verify the following chronological partitions. The split is preserved through preprocessing and modelling; no random re-split is performed.

Table 5.1: Documented chronological split

Partition	Period	Records
Training	January–August 2013	217,727
Validation	September–October 2013	55,628
Testing	November–December 2013	53,991
Total	January–December 2013	327,346

Training data is used to fit model parameters and preprocessing. Validation data supports model selection and threshold choices. Test data is held back for final evaluation. Chronological splitting better approximates predicting future observations than a random split.

5.2 Preprocessing

The Task 4 notebook defines 21 original predictors: four categorical features (carrier, origin, destination, route) and 17 numerical features (distance, cyclical schedule/calendar encodings, and lagged historical rates/counts). Numerical features are median-imputed and standardized; categorical features are imputed and one-hot encoded. The transformer is fitted on training data only and applied unchanged to validation and test. This produces 356 transformed columns. Missing historical rates remain for some early/no-history observations and are handled by the train-fitted imputation pipeline.

5.3 Reproducibility

Record the split dates, target definition, feature list, preprocessing configuration, random seeds where relevant, and software/library versions. Any historical aggregates must be computed in a time-safe manner.

Chapter 6

05 — Logistic Regression

6.1 Purpose

Logistic Regression is the interpretable baseline for the binary classification task. It estimates a class probability through the logistic function:

$$P(y = 1 \mid X) = \frac{1}{1 + \exp(-z)},$$

where z is a linear combination of input features and learned coefficients.

6.2 Training and Evaluation Protocol

Train Logistic Regression on the training partition after applying the documented preprocessing. Use validation data for any model selection or operating-threshold choice. Do not use the test set to tune the model. The notebook fits `LogisticRegression(max_iter=2000, class_weight="balanced", random_state=42, solver="lbfgs")`. Class weighting increases sensitivity to the minority delayed class. The validation-selected F1 threshold is 0.4721; the default 0.50 threshold is also retained for a baseline comparison.

6.3 Baseline Results

At the default threshold 0.50, the held-out test accuracy is 0.6329, precision 0.3344, recall 0.4736, F1 0.3920, ROC-AUC 0.6187, and PR-AUC 0.3387. With the threshold 0.4721 selected on validation for F1, test precision is 0.3248, recall 0.5402, F1 0.4057, and accuracy 0.6045. The tuned-threshold confusion matrix is [[25,346, 15,153], [6,203, 7,289]] in $[[TN, FP], [FN, TP]]$ order. The recall increase is accompanied by more false positives and lower accuracy, which is an expected operating-threshold trade-off.

Chapter 7

06 — XGBoost

7.1 Purpose

XGBoost is a gradient-boosted decision-tree method that can model nonlinear relationships and interactions among carrier, route, scheduled time, distance, and historical features.

7.2 Training Protocol

Use the same target definition, chronological partitions, and leakage-safe feature policy as the baseline to enable a fair comparison. The saved XGBoost notebook uses up to 800 estimators, max depth 5, learning rate 0.05, subsample 0.85, column sample 0.85, minimum child weight 5, L2 regularization 2.0, histogram tree construction, and a training-derived `scale_pos_weight`. Early stopping uses the validation set with patience 40; the best iteration is 82. The validation PR-AUC recorded at the best iteration is 0.2658.

7.3 Interpretability and Diagnostics

If available, include feature importance or SHAP-based analysis, precision–recall curves, confusion matrices, and error breakdowns. These outputs should be interpreted as model diagnostics rather than causal explanations.

7.4 XGBoost Results

At threshold 0.50, test accuracy is 0.6616, precision 0.3458, recall 0.3969, F1 0.3696, ROC-AUC 0.6288, and PR-AUC 0.3430. The validation-selected F1 threshold is 0.465672. Applying it to the held-out test set yields accuracy 0.6300, precision 0.3392, recall 0.5068, F1 0.4064, ROC-AUC 0.6288, and PR-AUC 0.3430. The tuned-threshold confusion matrix is $[[27{,}177, 13{,}322], [6{,}654, 6{,}838]]$ in $[[TN, FP], [FN, TP]]$ order. The threshold was selected on validation, not on the test set.

Chapter 8

07 — Evaluation and Model Comparison

8.1 Evaluation Metrics

Accuracy alone may be insufficient if the classes are imbalanced. Use the metrics actually computed in the notebooks and explain why they matter.

Table 8.1: Metric interpretation

Metric	Interpretation
Precision	Among flights predicted delayed, the fraction that were delayed.
Recall	Among actually delayed flights, the fraction identified.
F1-score	Harmonic mean of precision and recall.
ROC-AUC	Discrimination across classification thresholds.
PR-AUC	Precision–recall performance, especially useful for imbalanced targets.
Calibration	Agreement between predicted probabilities and observed event rates.
Recall at selected precision	Operational trade-off at a specified precision requirement.

8.2 Comparison Table

The table below uses the model notebooks' saved test outputs. The tuned operating thresholds were selected on validation data. ROC-AUC and PR-AUC are probability-ranking metrics and do not change merely because the classification threshold changes.

Table 8.2: Held-out test-set comparison from saved model notebook outputs

Model / cut-off	Precision	Recall	F1	ROC-AUC	PR-AUC
LR, 0.500	0.3344	0.4736	0.3920	0.6187	0.3387
LR tuned, 0.472	0.3248	0.5402	0.4057	0.6187	0.3387
XGB, 0.500	0.3458	0.3969	0.3696	0.6288	0.3430
XGB tuned, 0.466	0.3392	0.5068	0.4064	0.6288	0.3430

8.3 Interpretation

Under the F1-oriented threshold comparison, XGBoost has a marginally higher test F1 (0.4064 versus 0.4057), PR-AUC (0.3430 versus 0.3387), and ROC-AUC (0.6288 versus 0.6187). Logistic Regression has higher recall at its selected threshold (0.5402 versus 0.5068), while XGBoost has slightly higher precision (0.3392 versus 0.3248). The F1 difference is only 0.0007 and should not be overstated as a decisive win. Both models miss many delayed flights and produce many false alarms; model choice therefore depends on whether the operation prioritizes catching delays or limiting false alerts.

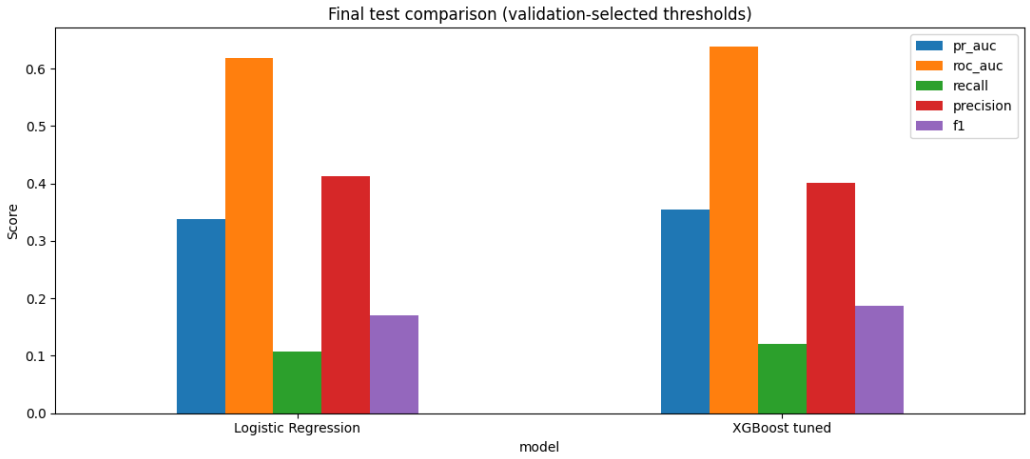


Figure 8.1: Additional model-optimization notebook comparison at an operational precision constraint. This is a separate calibrated-threshold experiment, not the main F1-threshold comparison above.

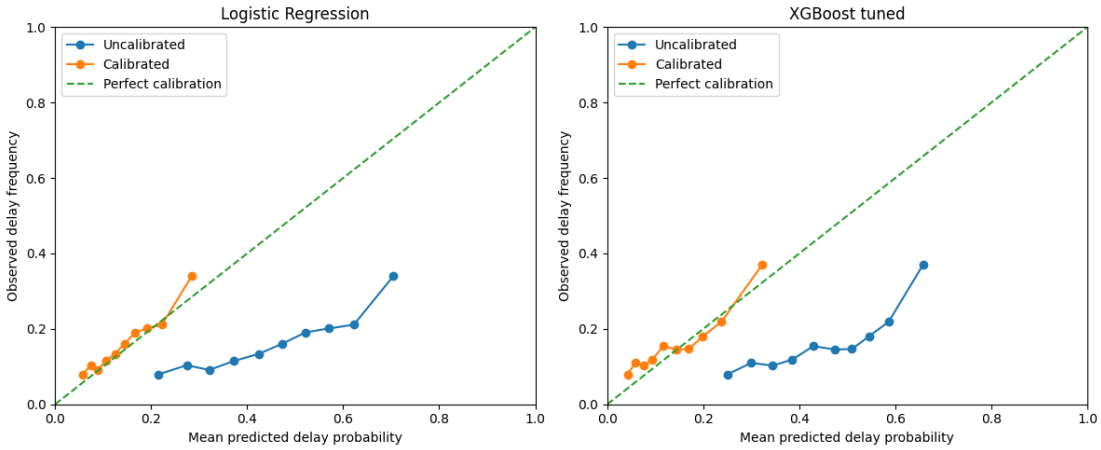


Figure 8.2: Validation calibration curves for the optimization notebook. Calibration adjusts probability reliability; it does not by itself improve ranking metrics.

8.4 Additional Calibration and Operational Threshold Experiment

The separate optimization notebook explores three XGBoost configurations and selects candidate 3 by tuning-set PR-AUC (0.2332; best iteration 101). It then fits Platt scaling on a dedicated calibration portion of validation data and chooses a threshold to satisfy a minimum validation precision target of 0.35. This target is an explicit operational assumption in the notebook, not a requirement established by the dataset. The selected thresholds are 0.2476 for Logistic Regression and 0.2572 for tuned XGBoost. On the final test set, the calibrated models have Brier scores of 0.1942 and 0.1919 respectively, and PR-AUC values of 0.3387 and 0.3553. At those thresholds, recall is only 0.1078 for Logistic Regression and 0.1215 for XGBoost, illustrating that a precision floor can sharply reduce the number of delayed flights identified. These results belong to a distinct experimental protocol and should not be mixed directly with the F1-threshold rows.

8.5 Conclusion

Flight Club builds a pre-departure binary classifier for arrival delays exceeding 15 minutes. The executed workflow starts with 336,776 raw flights, retains 327,346 completed flights with observed arrival outcomes, engineers 21 leakage-audited predictors, and transforms them into 356 model columns using training-only preprocessing. Chronological evaluation shows a challenging distribution shift: the delayed-flight rate falls to 15.05% in validation, then rises to 24.99% in test. Under validation-selected F1 thresholds, XGBoost reaches F1 0.4064 and Logistic Regression reaches 0.4057. The result is close: XGBoost has slightly stronger ranking metrics, while Logistic Regression identifies a larger share of delayed flights at its selected threshold. Neither model is sufficiently reliable to treat its output as a definitive operational decision without further validation, calibration monitoring, and a cost-based evaluation.

Limitations and Future Work

The documented dataset covers 2013 and may not represent current airline operations. Historical delay rates are proxies for prior behaviour rather than live congestion. Model validity depends on the prediction-time assumption and strict time-safe feature construction. The data covers one year (2013), so results may not transfer to current schedules, carriers, airport operations, or weather patterns. The sharp change in positive-class prevalence across time partitions indicates temporal non-stationarity and makes a single random split inappropriate. The historical feature engineering must ensure that only outcomes available before the prediction date contribute to each lagged statistic. The optimization notebook reports calibration and threshold experiments, but the precision-constrained thresholds identify relatively few delayed flights. Future work should evaluate multiple years and rolling-origin backtests, incorporate only weather/operational forecasts available at prediction time, monitor calibration and prevalence drift, compare against simple operational baselines, quantify false-positive/false-negative costs, and implement a documented rescheduling simulation. The repository title mentions rescheduling, but the notebooks in this ZIP primarily evaluate prediction; a rescheduling policy should not be presented as validated unless separately implemented and tested.

Team Contributions

The following allocation follows the project documentation and should be confirmed by the team before submission.

Member	Student ID	Documented responsibility
Khyati Batra	2401010229	Chronological split and preprocessing
Agrima Ojha	2401010047	XGBoost; evaluation, comparison, and final analysis
Khushi Jain	2401010227	Exploratory data analysis and target definition
Suhani Gupta	2410100463	Dataset understanding, data dictionary, leakage audit, and feature engineering
Akshat Aggrawal	2401010054	Logistic Regression baseline

References

1. Suhani Gupta et al. *T004-Flight-Club: Flight Delay Prediction and Rescheduling Using Gradient Boosting and Temporal Deep Learning*. Project repository. https://github.com/suhanigupta786/T004-Flight-Club

2. Suhani Gupta et al. *Data Dictionary: Flight Delay Prediction & Rescheduling*. Project repository. https://github.com/suhanigupta786/T004-Flight-Club/blob/main/Data_Dictionary.md

3. Kaggle. *Flights: 2013 US flight records*. Dataset source linked from the project README. https://www.kaggle.com/

4. U.S. Bureau of Transportation Statistics. *Airline On-Time Performance Data*. https://www.transtats.bts.gov/

Chapter A

Final Verification Checklist

- Verify dataset size and split counts against executed notebooks.
- EDA figures and interpretations inserted from saved notebook outputs.
- Confirm target missing-value handling and final feature list.
- Confirm no post-departure or target-derived fields enter the model.
- Add exact Logistic Regression and XGBoost configurations.
- Model metrics replaced with saved notebook validation/test results; check run versions before submission.
- Confusion matrices and calibration plots included where saved outputs were available.
- Confirm team roles and student IDs.
- Check the course rubric and formatting requirements before submission.